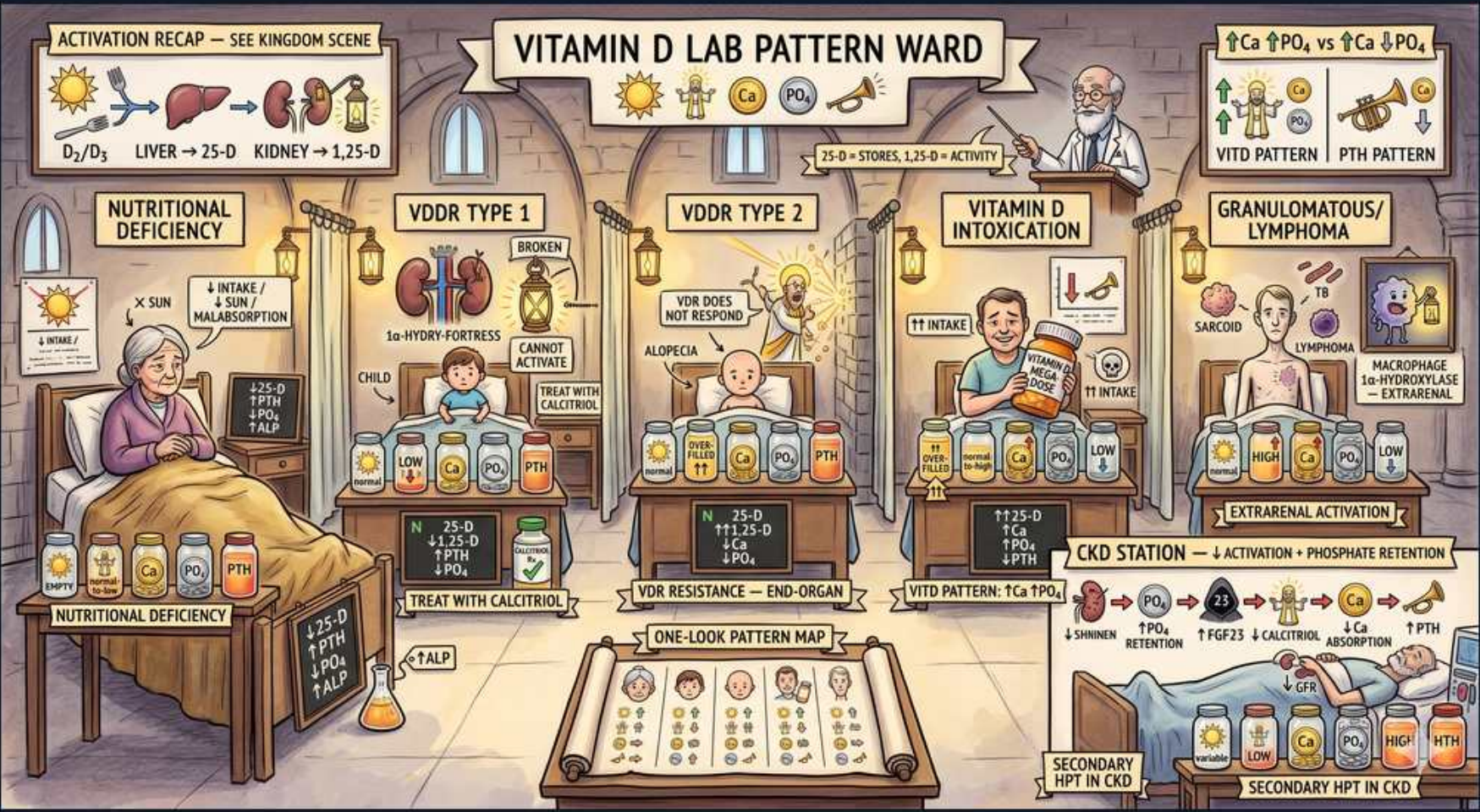


Calcium/PTH/VitD — The Vitamin D Lab Pattern Ward



1. Activation recap

WHAT YOU SEE

Top-left recap strip: a sun and pills feed D2/D3 into a LIVER organ stamped '25-D / STORES', then an arrow to a KIDNEY stamped '1,25-D / ACTIVITY' — the two-stop relay drawn left-to-right shows liver makes the storage form, kidney makes the active form.

WHAT IT ENCODES

Vitamin D activation is a two-hydroxylation relay. Skin (UVB on 7-dehydrocholesterol) and diet supply D3 (cholecalciferol) and D2 (ergocalciferol). The LIVER performs 25-hydroxylation (CYP2R1) → 25-OH-D (calcidiol), the STORAGE form and the best marker of vitamin D status (it has the long half-life, ~2–3 weeks, and large pool). The KIDNEY performs 1-alpha-hydroxylation (CYP27B1) → 1,25-(OH)2-D (calcitriol), the ACTIVE form. 1-alpha-hydroxylase is stimulated by PTH and low phosphate, and inhibited by FGF23, high calcium, and calcitriol itself. Mnemonic: 'Liver gives the 25 status; Kidney activates to 1,25.'

BOARD TRAP

WRONG ANSWER: 'Order 1,25-vitamin D to assess for vitamin D deficiency.' Discriminator: measure 25-OH-D for STATUS — calcitriol (1,25-D) has a short half-life and is tightly regulated, so it can be normal or even HIGH in early deficiency (driven up by secondary PTH). 1,25-D testing is reserved for specific situations (suspected CKD, VDDR, granulomatous disease, oncogenic osteomalacia).

2. Nutritional deficiency bed

WHAT YOU SEE

An elderly woman in a hospital bed (low sun/low intake) with bedside lab cards reading 25-D down, 1,25-D down, Ca/PO4 down, PTH UP, ALP UP — the frail housebound figure literally IS dietary/sunlight vitamin D deficiency with a compensatory high-PTH response.

WHAT IT ENCODES

Nutritional vitamin D deficiency (low sun exposure, low intake, malabsorption, housebound elderly): low 25-OH-D leads to reduced intestinal Ca and PO4 absorption → low/low-normal serum Ca and PO4 → compensatory secondary hyperparathyroidism (HIGH PTH), which raises calcitriol toward normal and elevates ALP (osteoblast activity from osteomalacia). Deficiency threshold: 25-OH-D <20 ng/mL is deficient, 20–30 insufficient. Treat with cholecalciferol (D3) repletion.

BOARD TRAP

WRONG ANSWER: 'High PTH means a parathyroid adenoma (primary hyperparathyroidism).' Discriminator: in vitamin D deficiency the high PTH is SECONDARY and APPROPRIATE — accompanied by LOW/normal calcium and LOW 25-OH-D. Primary HPT has HIGH calcium with high PTH. The calcium level and the low 25-OH-D distinguish secondary (deficiency) from primary HPT.

3. VDDR type 1 panel

WHAT YOU SEE

A child panel labelled VDDR TYPE 1 with a broken/disabled kidney enzyme and lab cards 25-D HIGH but 1,25-D LOW — the broken kidney machine is the missing 1-alpha-hydroxylase: substrate piles up, active form can't be made.

WHAT IT ENCODES

Vitamin D-dependent rickets type 1 (VDDR1, '1A') is an autosomal recessive deficiency of renal 1-alpha-hydroxylase (CYP27B1). The activation step fails: 25-OH-D is normal/HIGH but 1,25-D (calcitriol) is LOW. Result: hypocalcemia, secondary hyperPTH, hypophosphatemia, and rickets in infancy. Treatment is CALCITRIOL (active vitamin D) — bypassing the missing enzyme; plain vitamin D doesn't work.

BOARD TRAP

WRONG ANSWER: 'Treat VDDR1 with high-dose ergocalciferol/cholecalciferol.' Discriminator: the enzyme that ACTIVATES vitamin D (1-alpha-hydroxylase) is missing, so plain vitamin D can't be converted — give CALCITRIOL (or alfacalcidol). The clue is normal/high 25-D but LOW 1,25-D.

4. VDDR type 2 / VDR resistance

WHAT YOU SEE

A bald child (alopecia) labelled VDDR TYPE 2 beside a sign '1,25-D HIGH / VDR DOES NOT RESPOND' — the alopecia plus plenty-of-hormone-but-no-effect is the picture of a broken vitamin D RECEPTOR (end-organ resistance).

WHAT IT ENCODES

Vitamin D-dependent rickets type 2 (VDDR2, hereditary vitamin D-resistant rickets) is an autosomal recessive END-ORGAN resistance from a mutated vitamin D receptor (VDR). Calcitriol is made fine — in fact 1,25-D is HIGH (compensatory) — but tissues can't respond. Result: hypocalcemia, secondary hyperPTH, rickets, and characteristic ALOPECIA (often total). Treatment: very high-dose calcitriol and calcium; many require IV/high-dose oral calcium because the receptor defect can't be overcome.

BOARD TRAP

WRONG ANSWER: 'High 1,25-D rules out a vitamin D problem — the patient is fine.' Discriminator: in VDDR2 calcitriol is HIGH but the VDR can't respond (receptor defect), so giving more vitamin D fails. ALOPECIA is the key clue distinguishing VDDR2 (high 1,25-D + alopecia) from VDDR1 (low 1,25-D, no alopecia).

5. Vitamin D intoxication

WHAT YOU SEE

A man clutching pill/supplement bottles labelled INTOXICATION, with lab cards 25-D HIGH, Ca HIGH, PO4 HIGH, PTH LOW — the overflowing supplement bottles literally ARE vitamin D overdose driving up both calcium and phosphate while PTH is suppressed.

WHAT IT ENCODES

Vitamin D toxicity (excess supplements, megadosing): markedly HIGH 25-OH-D (often >100 ng/mL) saturates vitamin D-binding protein, freeing vitamin D metabolites that drive intestinal Ca AND PO4 absorption → HYPERcalcemia and HYPERphosphatemia with appropriately SUPPRESSED (low) PTH. Can cause nephrocalcinosis, AKI, and metastatic calcification. Treatment: stop vitamin D, hydration, and glucocorticoids if severe.

BOARD TRAP

WRONG ANSWER: 'High calcium with low PTH must be malignancy.' Discriminator: vitamin D intoxication raises BOTH Ca and PO4 (with low PTH and very high 25-D), whereas humoral hypercalcemia of malignancy (PTHrP) gives high Ca with LOW phosphate. The phosphate direction plus a sky-high 25-OH-D points to toxicity. Note it also differs from primary HPT (high Ca, LOW PO4, high PTH).

6. Granulomatous / lymphoma

WHAT YOU SEE

Sarcoid/lymphoma figures beside a MACROPHAGE that is running its own extra-renal 1-alpha-hydroxylase — the rogue macrophage cranking out 1,25-D off-site is the source of PTH-independent, steroid-responsive hypercalcemia.

WHAT IT ENCODES

Granulomatous diseases (sarcoidosis, TB, histoplasmosis, berylliosis) and some lymphomas cause hypercalcemia because activated MACROPHAGES express 1-alpha-hydroxylase EXTRA-RENALLY, converting 25-D → 1,25-D unchecked by normal feedback. Labs: HIGH 1,25-D, hypercalcemia, hypercalciuria, SUPPRESSED PTH, and often elevated ACE (sarcoid). Treatment is GLUCOCORTICOIDS (inhibit macrophage 1-alpha-hydroxylase) plus sun/vitamin D avoidance.

BOARD TRAP

WRONG ANSWER: 'Treat sarcoid hypercalcemia with more vitamin D or with bisphosphonates as first-line.' Discriminator: this is 1,25-D-mediated hypercalcemia with SUPPRESSED PTH — give GLUCOCORTICOIDS (which shut off macrophage 1-alpha-hydroxylase) and AVOID vitamin D/sun. Giving vitamin D worsens it. High 1,25-D with low PTH and low 25-D-needs points to extra-renal activation.

7. Extra-renal activation tag

WHAT YOU SEE

A banner reading EXTRA-RENAL ACTIVATION pointing at the macrophage — flags that this 1-alpha-hydroxylase sits OUTSIDE the kidney and so escapes the normal calcium/calcitriol feedback brakes.

WHAT IT ENCODES

Extra-renal 1-alpha-hydroxylase (in granuloma macrophages and some lymphoma cells) is NOT subject to the normal negative feedback that restrains the renal enzyme — it is not suppressed by high calcium or by calcitriol and is not driven by PTH. This unregulated calcitriol production is why granulomatous hypercalcemia is PTH-independent and corticosteroid-responsive.

BOARD TRAP

WRONG ANSWER: 'Calcitriol production is always tightly controlled, so 1,25-D can't be the cause of hypercalcemia.' Discriminator: extra-renal (macrophage) 1-alpha-hydroxylase escapes feedback regulation, so 1,25-D rises unchecked — measure 1,25-D (HIGH) with suppressed PTH to identify granulomatous/lymphoma-driven hypercalcemia.

8. Treat with calcitriol note

WHAT YOU SEE

A treatment banner under the VDDR1 panel reading TREAT WITH CALCITRIOL — the prescription tag marks the rule: when the kidney's activating enzyme is gone, you must hand over the already-active hormone, not plain vitamin D.

WHAT IT ENCODES

Conditions where the kidney's activation step is impaired require ACTIVE vitamin D (calcitriol or analogs like alfacalcidol/paricalcitol), not plain D2/D3: VDDR1 (absent 1-alpha-hydroxylase) and CKD (lost renal 1-alpha-hydroxylase). Giving plain cholecalciferol can't help if the activating enzyme is gone — you must supply the already-activated hormone.

BOARD TRAP

WRONG ANSWER: 'Plain vitamin D supplementation is sufficient in CKD and VDDR1.' Discriminator: when renal 1-alpha-hydroxylase is missing/impaired (CKD, VDDR1), plain vitamin D can't be activated — give CALCITRIOL or an active analog (paricalcitol in CKD to also help suppress secondary PTH).

9. VDR resistance / end-organ

WHAT YOU SEE

A banner under the VDDR2 panel reading VDR RESISTANCE — END-ORGAN — labels the defect as a downstream/receptor failure, so even abundant calcitriol can't get its signal through.

WHAT IT ENCODES

VDDR2 is end-organ (post-receptor) resistance: the vitamin D receptor itself is defective, so even abundant calcitriol can't signal. Supplementation alone often fails; management relies on very high-dose calcitriol and high-dose oral or IV calcium to force calcium balance independent of the receptor. Severity correlates with the degree of receptor dysfunction.

BOARD TRAP

WRONG ANSWER: 'Just give calcitriol — it always corrects vitamin D-dependent rickets.' Discriminator: in VDDR2 the receptor can't respond, so calcitriol alone (which works for VDDR1) is frequently INSUFFICIENT — high-dose calcium (sometimes IV) is required. Alopecia and HIGH 1,25-D flag VDDR2 over VDDR1.

10. CKD station

WHAT YOU SEE

A bottom-right CKD station with a damaged kidney showing low activation, PO4 retention, FGF23 UP, calcitriol DOWN — the broken-down kidney depot ties together lost 1-alpha-hydroxylase plus phosphate buildup driving secondary HPT.

WHAT IT ENCODES

CKD impairs vitamin D in two ways: loss of functional renal 1-alpha-hydroxylase (low calcitriol) and phosphate retention that raises FGF23 (which further suppresses 1-alpha-hydroxylase). Low calcitriol + hyperphosphatemia + hypocalcemia all drive secondary hyperparathyroidism. Management: phosphate binders, active vitamin D analogs (paricalcitol/calcitriol), and calcimimetics (cinacalcet) to control PTH.

BOARD TRAP

WRONG ANSWER: 'CKD hypocalcemia is best treated with plain vitamin D alone.' Discriminator: CKD loses the activating enzyme, so plain D can't be converted to calcitriol — use ACTIVE vitamin D (calcitriol/paricalcitol) and control phosphate first (giving active D with high phosphate risks raising the Ca×PO4 product and calcification).

11. One-look pattern map

WHAT YOU SEE

A bottom-center unrolled scroll/grid of icons mapping 25-D, 1,25-D, Ca, PO4, PTH across every disorder — the one-glance pattern board that says the diagnosis lives in the COMBINATION of arrows, never a single value.

WHAT IT ENCODES

The whole ward reduces to pattern recognition across 25-D, 1,25-D, Ca, PO4, and PTH. Anchors: Primary HPT = high Ca, low PO4, high PTH. Secondary HPT (deficiency/CKD) = low/normal Ca, high PTH, low 25-D or low calcitriol. Vitamin D toxicity = high Ca, high PO4, low PTH, very high 25-D. Granulomatous = high Ca, high 1,25-D, low PTH. PTHrP malignancy = high Ca, low PO4, low PTH, low PTHrP-positive. VDDR1 = low 1,25-D; VDDR2 = high 1,25-D + alopecia.

BOARD TRAP

WRONG ANSWER: 'You can diagnose calcium/vitamin D disorders from calcium alone.' Discriminator: the diagnosis comes from the COMBINATION — always pair serum calcium with PTH first, then add phosphate and the vitamin D metabolites. The same calcium can mean opposite diseases depending on whether PTH is high or suppressed and whether phosphate is high or low.

12. Secondary HPT in CKD

WHAT YOU SEE

A bottom banner SECONDARY HPT IN CKD with a patient figure and four glowing/enlarged parathyroids — the chronically over-stimulated glands picture how low calcitriol and retained phosphate keep PTH high in renal failure.

WHAT IT ENCODES

Secondary hyperparathyroidism in CKD: chronically low calcitriol, phosphate retention, and hypocalcemia continuously stimulate the parathyroids → HIGH PTH with LOW/normal calcium. Over years the glands hypertrophy and can become autonomous (tertiary HPT: persistently HIGH PTH now with HIGH calcium, often after transplant). Treatment escalates from binders/active D/calcimimetics to parathyroidectomy in refractory tertiary disease.

BOARD TRAP

WRONG ANSWER: 'High PTH with hypercalcemia in a CKD patient is primary hyperparathyroidism.' Discriminator: in long-standing CKD this is usually TERTIARY HPT (autonomous glands after years of secondary stimulation) — high PTH, now high Ca, in the renal-failure context. Secondary HPT (the earlier stage) has high PTH with LOW/normal calcium. The calcium level and CKD history separate secondary, tertiary, and primary.
